VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Armament

AFLCMC/WA

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1.0 Overview

An armament architecture integrates three key systems in order to enable an effective and lethal weapon system. These three systems are the Mission System, a Stores Management System (SMS), and stores. The armament architecture, described by the Agile Mission Suite (AMS) Government Reference Architecture (GRA), isolates certain aspects of this integration. This approach allows for stability in safety-critical elements that are unlikely to change–thus removing the need to re-certify those elements while still allowing rapid evolution in other Capabilities.

1.1 Purpose

The Armament Volume describes an Armament Integration Architecture (AIA) that can be used by a Mission System Integrator (MSI) during the development of a Mission System. This volume also describes the SMS Translator element which is required as part of a Mission System.

1.2 Scope

The Armament Volume aids MSIs in designing and implementing an AIA and the functions of the SMS Translator required element.

1.3 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or Minimum Procurable Unit (MPU) provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.3.1** for a list of informative references for the Armament Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	Software Architecture Volume	14.0

Table 1.3.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About Armament and Related Entities.

2.0 Armament

The warfighter requires the most up-to-date and capable technology to fight an agile, lethal, and effective enemy. **Figure 2.0.1** describes the AIA features provided to meet this goal.

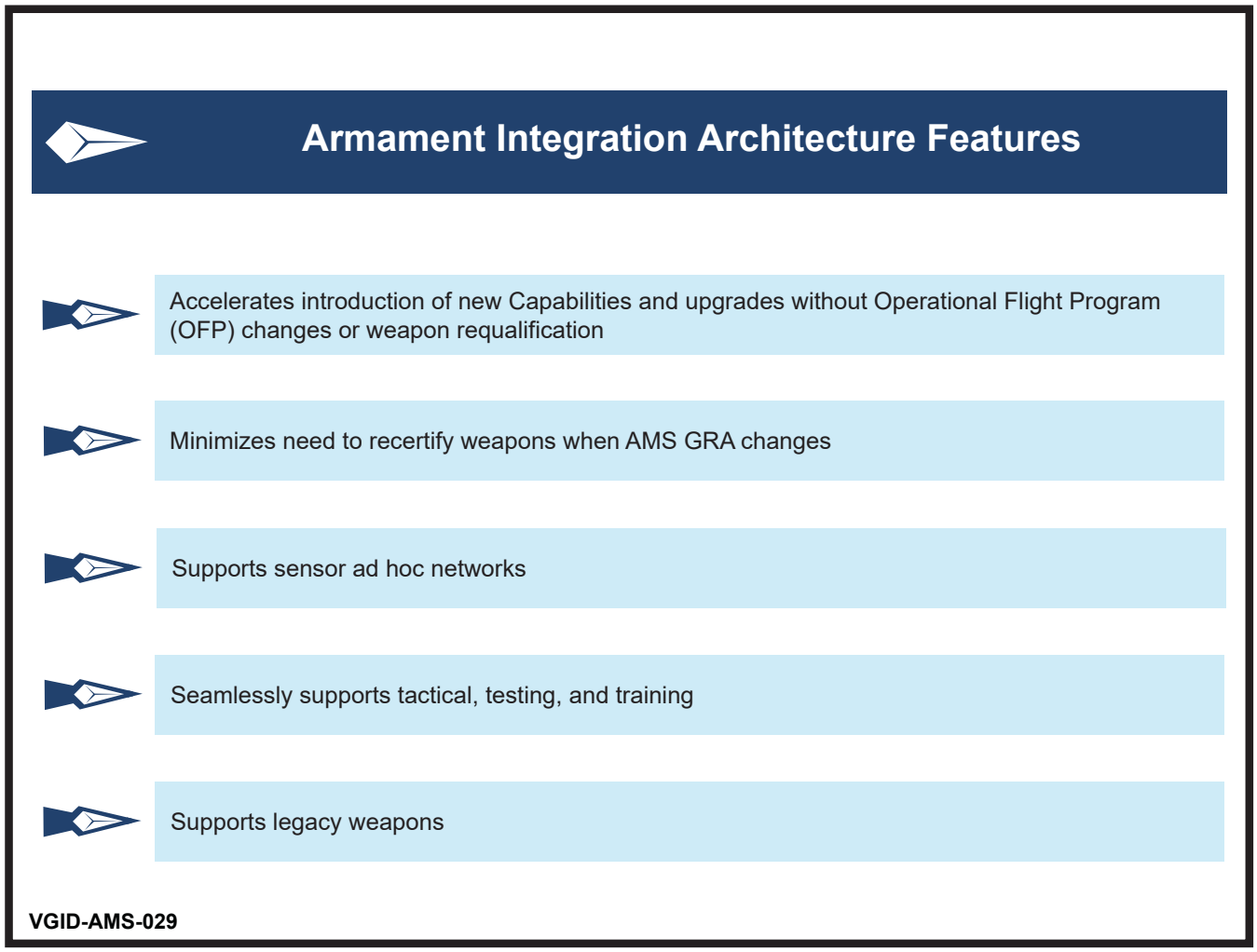


Figure 2.0.1: The AIA Features Enable Warfighters Access to Cutting-Edge Weapons Technology as it Develops.

2.1 Armament Integration Architecture

While the AIA contains areas outside of AMS GRA, examples of driving considerations for the design are:

- Division of responsibilities between Mission System, SMS, weapons/racks and other AIA components
- Interfaces between Mission System, SMS, and weapons/racks
- Cyber use cases (Note: does not include security and Anti-Tamper (AT))
- Execution behaviors of AIA components

The AIA contains three main areas:

- **SMS Translator Skill(s):** Translates AMS GRA Mission System Data to/from Store-specific SMS services via Weapon Systems Interface (WSI) (Inside AMS GRA boundary)
- **SMS platform Skill(s):** Translates non-AMS GRA Mission system data (including SMS Safety Critical Functions) to/from Store-specific SMS Services via WSI (outside AMS GRA boundary)
- **Store-specific SMS Services:** Software responsible for communicating with both stores and both Mission System and non-Mission System SMS Skills via WSI (outside AMS GRA boundary)

Splitting the AIA into three primary functions isolates the unique features of the store, passing that information in the standard messaging between the platform systems and the unique processing for the store and enabling rapid armament integration. Safety-Critical Functions (SCFs) are also isolated–minimizing their effect on the system.

The SMS Translator Skill is the only AIA area that is included in the AMS GRA boundary and is discussed in **Section 2.2**. The MSI implements the necessary SMS SCFs and Store-specific SMS Services.

2.2 Stores Management System (SMS) Translator

The SMS Translator translates all common mission system data from the platform to the common WSI data structure and communicates with the Store-specific SMS Services in the SMS. The platform sends Mission System data to the SMS Translator using Universal Command and Control Interface (UCI) messages in Open Mission Systems (OMS) or other Abstract Service Bus (ASB) or Avionics Backplane Bus (ABB) messages. The SMS Translator consists of any combination of AMS and/or OMS Services. The SMS Translator will leverage the open Platform Specific Interface (PSI) to communicate with stores via WSI. While communicating, the SMS Translator will not use definitions that are specific to any single weapon type. The SMS Translator does not implement SCFs. **Figure 2.2.1** and **Figure 2.2.2** show examples of patterns where a single SMS translator converts Mission System OMS data to WSI and sends it to the store-specific SMS services.

Attributes of the SMS Translator include:

- The SMS Translator Skill consists of AMS GRA and/or OMS Services.
- The Store-specific SMS Services facing interface is WSI compliant.

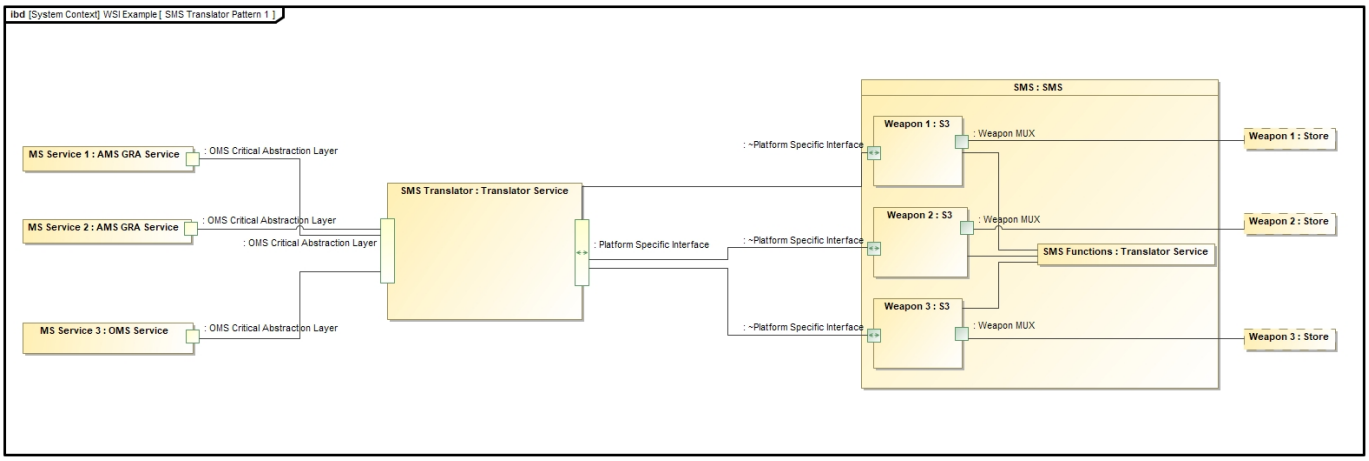


Figure 2.2.1: An SMS Translator Pattern Could Consist of One Translator Between Three Services and Three Weapons.

Figure 2.2.2 shows an additional pattern for a single SMS translator converting Mission System OMS data to WSI and sending it to the store-specific services. Additionally, it is not necessary that AMS GRA services be allocated to translator services in a 1:1 ratio as shown in the diagram.

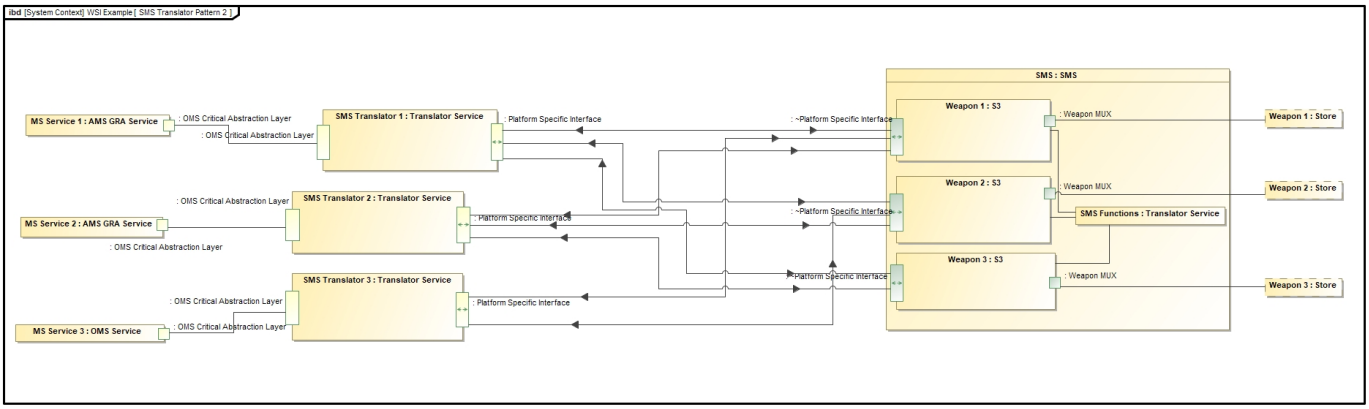


Figure 2.2.2: An SMS Translator Pattern Could Consist of Three Translators Between Three Services and Three Weapons.

3.0 Compliance

No requirements are contained in this Volume. It is the responsibility of the MSI to implement the SMS Translator Interface as specified above. See the Mission System Volume for the applicable requirements.

A.0 Acronyms

ABB	Avionics Backplane Bus
AIA	Armament Integration Architecture
AMS	Agile Mission Suite
ASB	Abstract Service Bus
AT	Anti-Tamper
GRA	Government Reference Architecture
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
OFP	Operational Flight Program
OMS	Open Mission Systems
PSI	Platform Specific Interface
SCF	Safety-Critical Function
SK	Starter Kit
SMS	Stores Management System
UCI	Universal Command and Control Interface
WSI	Weapon Systems Interface

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	Added a revision table to the volume	May 22, 2024	SK11.0
A	Created version	11/27/2024	SK11.1
B	This revision adds prose updates and clarifications.	July 24, 2025	SK13.0
C	This revision did not introduce any significant changes.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Armament Volume Between Documentation Releases.